Lang Shao

Cambridge, MA | shao.la@northeastern.edu
samueeelsiu.github.io | linkedin.com/in/lang-shao | github.com/samueeelsiu

EDUCATION:

Northeastern University Boston, MA
Candidate for Bachelor of Science Degree in Data Science and Mathematics *May* 2027
GPA: 3.73/4.0
Achievements: Dean's List
Relevant Coursework: Machine Learning and Data Mining, Database Design and Large-Scale Storage & Retrieval, Information Presentation and Visualization, Advanced Programming with Data, Foundations of Cybersecurity

EXPERIENCE:

Northeastern University (Structural Futures Lab, supervised by Prof. Demi Fang) Boston, MA

Data Analyst (Full-time Co-op for 6 months, then Part-time): *Jul 2025 - present*

- Built end-to-end data engineering and visualization workflows for building sustainability research, delivering production-ready geospatial datasets and interactive dashboards used in ongoing studies (see PROJECTS).
- Co-authored two research papers (see PUBLICATIONS), owning the statistical analysis, geospatial data pipelines, and survival modeling of building lifespans.

Beth Israel Deaconess Medical Center Boston, MA

Neuroimaging Research Co-op (Full-time, 6 months): *Jul 2024 - Dec 2024*

- Supported an NIH-funded neuroimaging study (Quantitative Susceptibility Mapping, Oxygen Extraction Fraction) by processing imaging data with Python on Linux-based pipelines; contributed to conference abstracts.
- Adapted open-source repositories for cerebral microbleed (CMB) detection; distilled technical literature into recurring team presentations on AI in medical imaging and segmentation methods.

PROJECTS:

Massachusetts Building Analysis Dataset & Dashboard | *Python, GeoPandas, Pandas, Scikit-learn, Plotly.js, D3.js*

- Built a geospatial data fusion pipeline matching 2M+ building polygons with point-level attribute data using multi-stage spatial joins and nearest-neighbor matching across Massachusetts.
- Integrated 5+ national and city-level datasets (e.g., USA Structures, NSI, Web Soil Survey, MassGIS) into a unified GeoPackage with occupancy conflict detection, soil property linkage, and material/foundation classification.
- Developed a Python preprocessor performing data cleaning, K-Means clustering and soil composition analysis, exporting chunked JSON for front-end consumption.
- Created a comprehensive interactive web dashboard with 20+ Plotly and D3 visualizations across multiple modules, covering building age, occupancy, height, gross floor area, material and foundation distributions, and data quality metrics.

Boston Building Lifespan Dashboard & Survival Analysis | *Python, Lifelines, PyCox, scikit-survival, Chart.js, Leaflet*

- Built a geospatial preprocessing pipeline in Python to spatially join 69K+ Boston building records with zoning district boundaries, computing lifespan metrics, material/foundation distributions, and gross floor area statistics for demolished vs. standing buildings.
- Developed an interactive single-page dashboard with 20+ Chart.js and Leaflet visualizations, including heatmaps, boxplots, scatter plots, and a zoning district deep-dive module comparing demolished and current building stock side-by-side.
- Applied 6+ survival analysis models (Kaplan-Meier, Nelson-Aalen, Cox PH, Weibull AFT, Random Survival Forest, DeepSurv) to predict building demolition probability.
- Identified overfitting in deep learning approaches at small event scales, validating traditional parametric models as more robust; findings contributed to a research publication (see PUBLICATIONS).

PUBLICATIONS:

- Melchiorre, Jonathan, **Lang Shao**, and Demi Fang. "Impact of Data Visualization on Sustainable Decision-Making in Structural Design." Under review. Preprint: [SSRN], Aug 2026.
- Melchiorre, Jonathan, **Lang Shao**, and Demi Fang. "The Life and Death of Buildings: A Survival Analysis of Urban Transformation in Greater Boston." In preparation.

SKILLS:

Programming: Python, R, SQL, JavaScript
Data & ML: Pandas, GeoPandas, Scikit-learn, PyTorch, Lifelines, scikit-survival, Survival Analysis
Visualization & Tools: D3.js, Plotly.js, Chart.js, Leaflet, Git, Linux, QGIS, Docker, Cassandra, Redis
Languages: Fluent in English, Native in Chinese and Cantonese **| Interests:** Music Production, Clarinet, E-sports, Soccer